DO NOW — RECALL

Programming paradigms: applied comparison

Name: _____ Class: _____ Date: _____

1. How does OOP differ from a procedural program?

2. What does a local variable mean?

3. If a field is private-set, how can callers change it safely?

4. Why does `Deposit(balance,5);` not update the caller's int balance if only a return value is produced?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

OOP groups state and behaviour into objects; procedural code is organised around procedures/functions.

2. Model answer

A variable accessible only within its declaration's scope.

3. Model answer

Through public methods/property logic defined by the class.

4. Model answer

The return is discarded and a value-type parameter is a copy; the caller must assign the returned result.

Vocabulary — know these terms

refactoring — Restructuring code while preserving expected behaviour.

encapsulation — Controlled access to internal state.

instance — Individual object created from a class.

return value — Output returned to the caller.

side effect — Observable change beyond a returned result.

NEW LEARNING

Read, annotate and explain

Procedural state

The caller passes an integer balance and explicitly stores the returned value.

Object state

Account object A stores Balance internally and Deposit changes that object's state.

Separate instances

Account B stays at 20 when Account A changes to 15; the objects have independent state.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Two accounts start at 8 and 20; only first account deposits 4. Trace both.

Predict

Account a's Balance becomes 12.

Model response

Account b remains at 20 because its state is separate.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Procedural AddFive(x) returns x+5. If caller x=2 discards the result, what is x?

Question 2

Two objects have balances A=10, B=30. A.Deposit(7) and B.Deposit(2) run. Find both balances and describe why they differ.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare one maintenance advantage and one risk of procedural and OOP implementations for a library system.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Calling a C# function that returns a new balance automatically changes the caller's original int variable.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.